

CONTACT

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[Zenodo record list](#)

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DETAILS

Location: Pamplona, Spain

Nationality: Nigerian

EDUCATION

2025 to expected Jun/Aug 2027

Erasmus Mundus Joint Master's Degree (EMJMD)

Cutting-Edge Technologies for Sustainable Agriculture (SUSTAGRI)

 Public University of Navarre, Spain (Sep 2025 to Jan 2026)

 UTAD, Portugal (Feb 2026 to Jun 2026)

 Agricultural University of Athens, Greece (Sep 2026 to Jan 2027)

2019 to 2024

B.Eng. Agricultural and Bioresources Engineering

First Class Honours, CGPA 4.77/5.00, Top 1 percent

 Ahmadu Bello University, Nigeria

2016 to 2018

National Diploma, Agricultural Engineering Technology

Distinction, CGPA 3.78/4.00, Top 1 percent

 Ahmadu Bello University, Nigeria

SKILLS

Python / Deep Learning

GIS and Remote Sensing

Computer Vision

Geostatistics

Agric. Machinery Systems

MATLAB / R

LANGUAGES

English (fluent)

MEMBERSHIPS

ASABE, NIAE, NSE, Pupil Engineer
COREN

Naziru HALILU

DIGITAL AGRICULTURE

PROFILE

Agricultural engineer with a long term research focus on the digitalization of agriculture, spanning computer vision, geospatial machine learning, agricultural control systems, and high throughput phenotyping, grounded in a strong foundation in agricultural machinery, mechanization, and farm power systems.

EMPLOYMENT HISTORY

EMJMD Research Scholar (Mobility Track)

Sep 2026 to Jan 2027

 Agricultural University of Athens, Greece

- Term 3 focus: efficient water use, smart irrigation (30 ECTS).
- Continued graduate-level research training in smart design and management of irrigation systems.
- Built NaCROP, a modular Python/Tkinter pipeline for evapotranspiration, soil-water balance, irrigation scheduling, and a four-step crop-growth simulation for five smallholder crops around Zaria, Nigeria. [Data](#)

Summer Intern, Computer Vision Group

Jun to Aug 2026

 Dept. of Computer Engineering, Ahmadu Bello University, Zaria

- Accomplished cross-condition transfer of crop disease recognition models, as measured by 64.4 percent classification accuracy (macro AUC 0.882 against a 5.3 percent chance baseline) across 19 disease classes on 1,543 field images, by extending the Geometry Agnostic Contrastive Learning (GACL) framework with four new geometry-invariant neural components. First author paper under review at Biosystems Engineering, Elsevier. [Data](#)

EMJMD Research Scholar (Mobility Track)

Feb to Jun 2026

 UTAD, Portugal

- Term 2 focus: precision agriculture for sustainable crop production (30 ECTS).
- Accomplished cross-system precision agriculture prediction and input cost reduction, as measured by R squared between 0.978 and 0.995 for grape ripening and maize yield regressions and a 2.69 tonne cut in lime demand. [Data](#)
- Accomplished more reliable soil nutrient and yield mapping via Multitask Singularity Regression Kriging (MSRK), as measured by a roughly 20 percent reduction in residual kriging uncertainty across 17,864 samples. [Data](#)
- Accomplished predictive harvest date forecasting for grape ripening, as measured by R squared of 0.9784, at a six-hectare UTAD vineyard. [Data](#)
- Accomplished a costed, spatially targeted pest management programme for grape moth across a 42.97-hectare Douro vineyard estate. [Data](#)

EMJMD Research Scholar (Mobility Track)

Sep 2025 to Jan 2026

 Public University of Navarre, Spain

- Term 1 focus: technological bases for agricultural sustainability (30 ECTS).
- Accomplished real-time, field-deployable crop health screening, as measured by 33.8 percent balanced accuracy against a 4.55 percent chance baseline, by building a fourteen-module RGB/multispectral vegetation indexing pipeline. [Data](#)
- Accomplished radiometrically consistent multi-temporal vegetation monitoring, as measured by a 39 and 31 percent reduction in temporal NDVI variance, by designing STADOS, a five-stage adaptive dark object subtraction framework. [Data](#)
- Accomplished accurate land suitability mapping for livestock slurry application, as measured by RMSE of 7.78 and R squared of 0.77 across 14.35 square kilometres near Tudela, Spain. [Data](#)
- Accomplished a 47.8 percent reduction in annual soil loss via a Bayesian-grounded digital twin for gully erosion biocontrol, hold-out R squared of 0.90. [Data](#)

- Built a six-notebook vector/raster GIS and Sentinel-1/2 remote sensing course package in Python and Google Earth Engine. [Data](#)

Undergraduate Project Assistantships

Mar 2024 to Mar
2025

 Ahmadu Bello University, Nigeria

- Contributed data collection, statistical analysis, and literature review across concurrent research projects in water and irrigation systems, soil and erosion assessment, and postharvest and animal systems.

Project Assistant, Remote Data Lead

2024 to 2025

 IAR-ABU, Nigeria

- Engineered adaptive irrigation scheduling algorithms fusing real-time soil moisture and microclimate telemetry for automated greenhouse control.

Student Research Assistant, Precision Crop Systems

2022 to 2024

 NAERLS, Nigeria

- Trained and validated CNN crop-stress detection models reaching 78 percent accuracy on multispectral UAV imagery.

Assistant Farm Manager (concurrent, part-time)

2022 to 2024

Yankari Nasarawa Farms, Nigeria

- Sustained animal welfare across a commercial poultry facility by managing automated climate-control and water-dosing systems.

Project Assistant, Smart Irrigation and Water Systems

2021 to 2022

BOMO-Basin Project, Nigeria

- Implemented closed-loop control logic for a one-hectare smart drip-irrigation array.

Research Intern

2017 to 2020

 NCAM, Ilorin and IAR, Zaria, Nigeria

- Built Python data-logging pipelines to benchmark mechanization efficiency; supported baseline hydrological research.

PUBLICATIONS

- Halilu, N., and Sulaiman, J. (2026). Real Time RGB Proxy Vegetation Indexing and Texture Analysis for UAV and Handheld Crop Imagery. IJRSI, 13(7), 2951 to 2989. [DOI](#)
- Halilu, N. (2026). A Hybrid CNN-BiLSTM-Attention Deep Learning Framework for Sediment Transport in an Agricultural Gully System. IJRSI, 13(7), 2920 to 2950. [DOI](#)
- Halilu, N., and Sulaiman, J. (2026). Sensing the Invisible: A Critical Review of NIR and Hyperspectral Analysis for Crop and Food Quality Assessment. IJRSI, 13(7), 2990 to 3020.
- Ismail, K., and Halilu, N. (2024). AI Driven Precision Agriculture System for Optimizing Maize Yield in Northern Nigeria. Proc. AfCPA, 2(1), 100 to 120.
- Auwal, M., and Yusuf, A., Naziru, H. (2020). Physio-chemical and bacteriological quality of water for irrigation purpose. Nigerian Research Journal of Engineering and Environmental Sciences, 5(1), 198 to 204.
- Halilu, N., and Sulaiman, J. Geometry Agnostic Contrastive Learning (GACL). Biosystems Engineering, Elsevier. Under review. [DOI](#)
- Halilu, N. (2026). Digital Twin for the Evaluation of Experimental Gully Biocontrol Using Morning Glory (Ipomoea spp.). Environmental Monitoring and Assessment, Springer Nature. Under review. [DOI](#)
- Halilu, N., and Sulaiman, J. Operationalizing GIS and Machine Learning across Contrasting Cropping Systems. European Journal of Agronomy, Elsevier. Under review. [DOI](#)
- Halilu, N. (2026). Multitask Singularity Regression Kriging for Spatio Temporal Soil NPK Crop Stress Yield Prediction. Zenodo Preprint. [DOI](#)
- Halilu, N. (2026). STADOS: Spectral Temporal Adaptive Dark Object Subtraction. Zenodo. [DOI](#)
- Halilu, N. (2025). A Systematic Review of IoT, Edge Computing, Time Series Modelling and Control Systems in Precision Irrigation. Zenodo Preprint.

AWARDS AND HONORS

Erasmus Mundus Joint Master Scholarship, 2025 · Best Graduating Student, B.Eng, 2024 · Nigerian Federal Government Award for Academic Excellence, 2024 · Best Undergraduate Research Paper Award, ABU, 2024 · Best Overall Graduating Student, National Diploma, 2019 · PTFD Undergraduate Fellow, 2023.